

SELECTED ENGINEERING PROJECTS

These projects reflect distinct engineering capabilities developed across my work, including reusable components built for Meridux

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ENGINEERING PHILOSOPHY

My approach to automation begins with understanding the operational system rather than selecting automation tools. Every solution is designed by analysing how information moves through an organisation, identifying structural bottlenecks, and engineering reliable workflows that improve operational performance while remaining maintainable over time.

Instead of automating isolated tasks, I architect complete workflow systems that integrate business rules, APIs, AI-assisted decision support, monitoring, reporting, and governance into a cohesive operational platform.

- Reliability before complexity
 - Data integrity before automation speed
 - Business objectives before technology selection
- Modularity before customisation
 - Operational visibility before optimisation

CORE TECHNICAL CAPABILITIES

WORKFLOW ENGINEERING

n8n

Workflow Orchestration

Business Process Automation

AI

OpenAI

Claude

Gemini

Prompt Engineering

PROGRAMMING

JavaScript

Python

INTEGRATION

REST APIs

Webhooks

Google Workspace

ENGINEERING

Solution Architecture

Modular Design

Error Handling

Monitoring

Audit Logging

Workflow Governance

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AI-Powered Patient Workflow Automation Platform (Meridux PatientCore)

Business Challenge

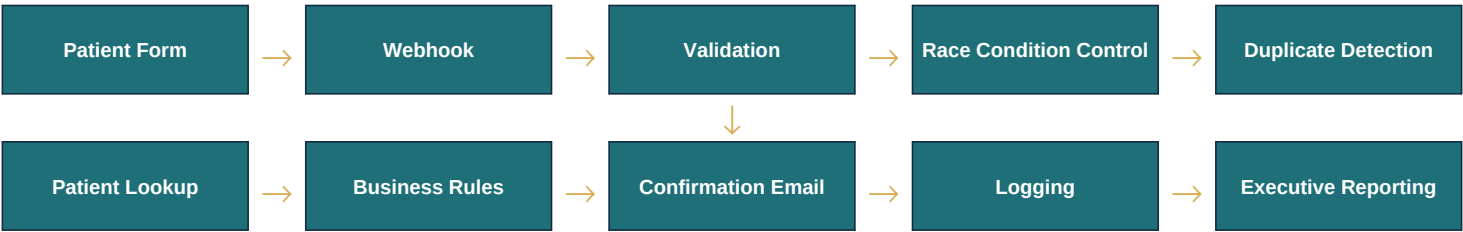
Healthcare providers often depend on manual administrative processes for handling patient enquiries, appointment requests, confirmations, duplicate checking, reporting, and follow-up activities. These fragmented workflows increase staff workload, introduce avoidable errors, delay response times, and reduce operational visibility.

The objective was to engineer a workflow architecture capable of automating the complete patient intake process while maintaining data accuracy, reliability, and operational transparency.

Design Objectives

- Eliminate repetitive administrative work
- Improve data consistency
- Prevent duplicate submissions
- Automate patient communication
- Generate operational reporting
- Build reusable workflow components
- Create an architecture suitable for future expansion

System Architecture



Engineering Decisions

Rather than storing workflow state within multiple disconnected applications, the architecture centralises operational data while separating validation, routing, reporting, and communication into independent workflow components.

This modular structure improves maintainability and enables future enhancements without redesigning the complete system.

Engineering Highlights

Workflow Orchestration

Designed a ten-stage workflow that automates patient processing from initial submission through confirmation and reporting.

Race-Condition Prevention

Engineered concurrency controls that prevent simultaneous submissions from generating duplicate workflow execution.

Intelligent Routing

Business rules automatically distinguish between new and returning patients, enabling conditional workflow execution.

Error Handling

Structured validation and monitoring ensure incomplete or inconsistent submissions are safely isolated without interrupting the primary workflow.

Operational Reporting

Daily executive summaries provide management with accurate operational insight without requiring manual reporting activities.

Technology Stack

n8n	JavaScript	REST APIs	Google Workspace
Google Sheets	Gmail API	OpenAI	Webhooks

Business Impact

The completed platform demonstrates how engineering principles can replace fragmented manual administration with an intelligent operational system that improves efficiency, reporting, and long-term maintainability.

Prospect Intelligence Engine (PIE)

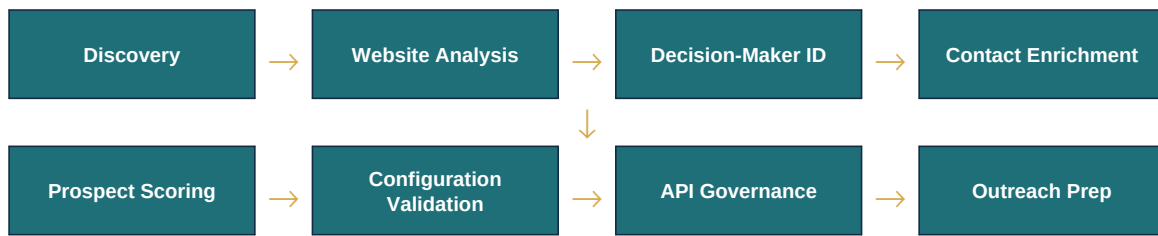
Business Challenge

Sales teams frequently rely on fragmented manual research when identifying prospects, collecting contact information, evaluating opportunity quality, and preparing outreach activities. This manual approach limits scalability and introduces inconsistencies throughout the prospecting process.

Design Objectives

- Automate business discovery
- Standardise prospect evaluation
- Improve data quality
- Reduce repetitive research
- Build reusable automation modules
- Support future expansion

Architectural Overview



Engineering Highlights

Modular Architecture

Each functional capability operates independently while contributing to a unified automation platform.

Configuration-Driven Design

Workflow behaviour is controlled through structured configuration rather than embedded logic, simplifying maintenance.

API Governance

Quota monitoring prevents service interruption while improving workflow reliability.

Data Integrity

Duplicate prevention and validation improve information quality before downstream processing.

Scalability

The architecture supports the addition of future workflow modules without affecting existing functionality.

Technology Stack

n8n	Google Places API	REST APIs	JavaScript
Google Sheets			

Multi-System API Integration Platform

Overview

Designed a reusable integration framework that synchronises information across multiple cloud services using APIs and webhooks. This architecture acts as a centralised data hub, substantially reducing manual data transfer and improving operational consistency across distributed systems.

Key Features

REST API Integration & HTTP Handling

Seamlessly connects distinct platforms by managing intricate HTTP requests and parsing external REST endpoints.

Authentication Management

Securely handles OAuth and API token lifecycles to maintain trusted connections with multiple service providers.

Error Handling & Retry Mechanisms

Implements robust fault-tolerance protocols, automatically retrying failed executions and managing timeout errors gracefully.

Webhook Processing

Captures real-time system events to initiate instant data synchronisations across the network.

JSON Transformation

Dynamically parses and transforms incoming JSON payloads to ensure data compatibility between disparate services.

Logging and Monitoring

Provides comprehensive execution logs to track system health and trace data pathways for rapid debugging.

Technology Stack

n8n	REST APIs	HTTP	JSON
OAuth	Google Workspace		

Implementation & Role

Engineered the comprehensive API integrations and architected the underlying workflow logic to ensure seamless data flow. Focused extensively on system reliability, stress-testing the authentication and retry mechanisms to prevent data loss, and documented the complete integration processes to facilitate future scaling and maintenance of the framework.

AI-Powered Business Process Automation

Overview

Developed AI-assisted workflows that automate repetitive business processes, reduce manual effort, and improve decision-making by combining workflow automation with large language models.

Key Features

AI-Assisted Content Generation

Leverages large language models to dynamically create high-quality content.

Automated Document Creation

Dynamically compiles and generates documents based on AI-processed inputs.

Workflow Orchestration

Seamlessly directs logic and data between various tools and APIs.

Prompt Engineering

Utilises highly tuned system prompts to ensure accurate, predictable AI outputs.

Intelligent Data Processing

Analyses and extracts insights from unstructured data for downstream systems.

Email Automation

Drafts, summarises, and schedules intelligent email communications automatically.

Business Process Optimisation

Identifies operational bottlenecks and resolves them via continuous automated action.

Technology Stack

n8n	ChatGPT	OpenAI API	REST APIs
Google Workspace			

Implementation & Role

Designed the core automation workflows and integrated the various AI services required to power the system. A significant portion of the work involved optimising AI prompts to guarantee robust and reliable outputs, extensively validating data transformations, and actively monitoring workflow performance to maintain peak efficiency.

Workflow Monitoring & Error Management Framework

Overview

Developed a reusable workflow monitoring and error management framework to improve the reliability and maintainability of automation systems. This architecture guarantees that failures are gracefully handled, properly logged, and instantly reported to administrators, ensuring minimal downtime for critical business operations.

Key Features

Centralized Error Handling

Intercepts and manages faults across all workflows through a unified, globally accessible error management protocol.

Failure Notifications

Dispatches immediate alerts via email and instant messaging to stakeholders the moment an anomaly is detected.

Validation Checks

Enforces strict data validation prior to critical processing steps, mitigating bad data entry and logic breaks.

Modular Monitoring Components

Built as a plug-and-play framework, easily deployable to any new or existing n8n workflows.

Automated Execution Logging

Records detailed state data for each process iteration, creating an irrefutable audit trail of system activities.

Retry Logic

Integrates sophisticated fall-back mechanisms to automatically retry operations on transient failures before escalating.

Diagnostic Reporting

Aggregates fault data to provide actionable diagnostic reports, speeding up the debugging process.

Technology Stack

n8n	Google Sheets	Gmail	HTTP Requests
JSON			

Implementation & Role

Took full ownership of the monitoring architecture design, implementing the intricate error-handling workflows and establishing the foundational logging protocols. Performed end-to-end testing under high-failure scenarios and documented the framework to facilitate easy onboarding and future maintenance by other developers.

Designing Reliable AI Automation Systems

Every workflow I design follows a consistent engineering methodology.

1

Understand the System Before the Technology

Automation should be based on operational requirements rather than available software tools.

2

Design for Reliability

Every workflow should continue operating predictably under normal and exceptional conditions.

3

Preserve Data Integrity

Validation, duplicate prevention, concurrency control, and audit logging protect the quality of operational data.

4

Build Modular Architectures

Reusable workflow components simplify maintenance and accelerate future development.

5

Prioritise Operational Visibility

Automated reporting, monitoring, and structured logging enable organisations to understand system performance without manual investigation.

6

Engineer for Business Outcomes

Technology is valuable only when it improves efficiency, reliability, decision-making, or customer experience.

My work combines systems engineering, software integration, and AI-powered workflow automation to transform complex operational processes into reliable, scalable business systems. I approach every solution as an architectural challenge, ensuring that technology serves clearly defined business objectives rather than becoming an objective in itself.